

MLOps & Deployment - Brief Notes

1. Docker Basics

Docker is a containerization platform that packages an application and its dependencies into containers, ensuring it runs consistently across different environments.

Key concepts: Image, Container, Dockerfile, Docker Hub.

Benefits: Portability, consistency, easy deployment.

2. FastAPI

FastAPI is a modern, high-performance Python framework for building REST APIs. It provides automatic API documentation, validation, and is widely used for serving ML models.

3. REST API Creation

REST APIs allow applications to communicate over HTTP using methods like GET (retrieve), POST (create/predict), PUT (update), and DELETE (remove). ML models are commonly exposed through a POST /predict endpoint.

4. ML Model Serialization

Serialization saves trained models for later use. Pickle is a general-purpose serializer, while Joblib is optimized for NumPy arrays and Scikit-learn models.

5. GitHub Actions

GitHub Actions automates workflows such as testing, building Docker images, and deploying applications whenever code is pushed to GitHub.

6. CI/CD Concepts

Continuous Integration (CI) automatically builds and tests code changes. Continuous Delivery/Deployment (CD) automatically releases successful builds to production, enabling faster and more reliable deployments.

7. MLflow Overview

MLflow manages the machine learning lifecycle by tracking experiments, logging parameters and metrics, storing models, and managing model versions.

8. Cloud Deployment (AWS/Azure)

Deploy ML applications on cloud platforms using services such as AWS EC2, S3, ECS/EKS, SageMaker or Azure VMs, Blob Storage, AKS, and Azure Machine Learning. Typical flow: Train → Save → FastAPI → Docker → Cloud Deployment.

9. Final Project Presentation

Present the problem statement, dataset, preprocessing, model, evaluation metrics, deployment architecture, live demo, and future improvements.

Typical MLOps Workflow

Collect Data → Preprocess → Train Model → Evaluate → Save Model → FastAPI API → Docker →
GitHub → GitHub Actions (CI/CD) → Cloud Deployment → Monitor (MLflow).